

# Topología inicial

**Definición 1 (Topología inicial).** Sean  $X \neq \emptyset$ ,  $\{(Y_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$  un conjunto de espacios topológicos y  $\mathcal{F} = \{f_\alpha: X \rightarrow Y_\alpha\}_{\alpha \in I}$ ,  $\sigma(X, \mathcal{F})$  es la topología inicial en  $X$  inducida por  $\mathcal{F} \iff$  es la topología menos fina que hace continuas todas las  $f_\alpha \in \mathcal{F}$ .

Como la intersección de topologías es una topología, tenemos que

$$\sigma(X, \mathcal{F}) := \bigcap \{ \mathcal{T} \text{ topología en } X : \forall \alpha \in I : f_\alpha: (X, \mathcal{T}) \rightarrow (Y_\alpha, \mathcal{T}_\alpha) \text{ es continua} \}.$$

## Referenciado en

- prop-base-topologia-inicial
- teo-abiertos-topologia-inicial
- lem-carac-continuidad-topologia-inicial
- topologia-debil
- prop-carac-convergencia-topologia-inicial
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